# Towards Understanding Information Encountering on the Web

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**Abstract**

This paper provides a preliminary report of a multi-phase research study currently in progress about accidental discovery of information (information encountering) on the Web. We first provide an introductory overview of the concept of information encountering and its research, and then we explore this concept as it pertains to electronic environments. Findings from two small-scale empirical studies pertaining to the characteristics of information encountering on the Web are then summarized and explained. In the conclusion of the paper we provide an initial interpretation of research findings and propose that (1) information users' capability to encounter information relates to the level of their sensitivity to the information environment and that (2) information encountering is a habitual activity adjusted to the unique characteristics of each information environment.

## INTRODUCTION

Hyper-links and an abundance of information make the World Wide Web (Web) a "promised land" for information users, especially for the accidental discovery of information or information encountering. While, there is a growing body of research on how users purposefully seek information on the Web (e.g., Tausher and Greenberg, 1997; Huberman et al. 1998; Choo at al., 1999), our understanding of how and why users find information on the Web by accident is still very limited. The purpose of the research activities presented in this paper is to fill this void and to help us develop a more complete understanding of users information behavior on the Web. It is our hope that an improved understanding of users' realistic experiences with information on the Web will inform the design of user interfaces, improve the overall usability on the Web, and subsequently increase the users' satisfaction with the Web as an information resource.

This paper synthesizes preliminary findings of research being done to investigate information encountering on the Web. This current work builds upon earlier research on the general concept of information encountering (Erdelez, 1995; 1996). Our current research is comprised of several discrete but complementary small-scale studies that did not follow a sequential and pre-determined design but rather evolved in response to the questions raised and the knowledge gained during the research process. ,

In this paper we will begin with an overview of the general concept of information encountering and its research. Then we will explore this concept as it specifically pertains to electronic environments. Next, we will briefly describe the methodology and key findings from two studies that are part of our research on information encountering on the Web. And, finally, we will offer initial interpretations of our research findings and introduce two propositions for understanding information encountering on the Web.

## BACKGROUND

Information behavior has been traditionally modeled in the literature as a process that leads the user from identification of a problem and an information need, through information seeking to location of information needed. Only recently more iterative models of information behavior have emerged (Bates, 1989; Kuhlthau, 1993) as well as models that include passive aspects of information behavior (Wilson, 1997; Williamson, 1998). These models Portray a more holistic picture of information behavior and provide a place for alternative types of information acquisition, such as accidental discovery of information.

In a very broad sense, information acquisition is an element of information behavior that can be categorized according to the various levels of (1) the user's active involvement in the information acquisition process, (2) the presence of a goal or a focus in the user's information acquisition activities, and (3) the formalization of information

collection strategies. Historically, the focus of information user research has been an active, problem-solving approach to information acquisition (Dervin, 1980; Mick et al., 1980; Belkin et al. 1982; Krikelas, 1983; Rouse & Rouse, 1984; Brown, 1991; Allen, 1996).

Until very recently accidental discovery of information has not traditionally been major part of information user studies research. In 1992 two independent research studies recognized that accidental discovery of information is a type of information acquisition experienced in information behavior in academic environments. Reneker (1992) concluded that the majority of information seeking done by academics is triggered by an expressed information need, but also recognized there is a "prevalence of latent information needs, which informants articulated when information or information-obtaining opportunities surface." (p. 123) She suggested that such opportunities often, emerge through`"chance encounter[s]" with readily accessible sources, such as newspapers, colleagues, supervisors, friends, and files created by users. Zhang (1992) directly addressed the context and characteristics of accidental discovery of information in a study of information-seeking patterns and behaviors of 468 undergraduate students at a Chinese university, suggesting that accidental discovery of information "is an unavoidable and healthy part of our information activities and that it is in some ways related to intentional information seeking." (p. 138)

Williamson (1998) conducted a study of information seeking behavior of 202 older adults in Australia and concluded that respondents exhibited "incidental information acquisition" in the context of routine activities (watching TV, listening to the radio, talking to other people). Incidental information acquisition, a synonym for accidental discovery of information, occurs unexpectedly as people engage in other activities. Williamson reported that study respondents "often 'picked up' information through these sources - information that they had not even known that they needed until they heard or read it." (p 25)

Erdelez (1995) surveyed 132 respondents in an academic setting about their accidental discovery of information. Based on this study Erdelez introduced the term *information encountering* and defined it as a memorable experience of an unexpected discovery of useful or interesting information. According to this definition users do not expect to find encountered information; this information could be useful to address either some specific problem or could be related to some area of the users' general interest. An additional characteristic of information encountering is that it can be described as an incident or event that usually intersects with some other information acquisition-related activity such as information seeking and browsing. However, information encountering can be also experienced during any other type of human activity that only broadly involves information behavior (e.g., while making copies at a photo copy machine). Erdelez also suggested that based on a user's attitude towards information encountering and the frequency of information encountering experiences, an information user could be identified as a super-encounterer, encounterer, occasional encounterer or a non-encounterer. Erdelez conducted interviews with 12 of the respondents who were selected from the study pool as super-encounterers. They were considered super-encounterers because they very frequently found information by accident; they relied on it, and considered it as an integral element of their information behavior. In a subsequent works, Erdelez discussed a conceptual model for exploration of an information encountering experience (Erdelez, 1997, 1999) with a four-dimensional analysis that included information user, information environment, information encountered and information needs addressed by encountering. This framework, she found, can be used to describe information encountering in various information environments as well.

In a follow-up analysis of data collected in 1995, Erdelez (1996) focused on the characteristics of information encountering on the Internet. She reported that 21% (26) of the 121[l] information encountering events that were re-analyzed occurred on the Internet, with half of them (13) occurring in the Web. Yet, interviews with the 12 "super-encounterers" revealed a very low interest in information encountering on the Internet. Only eight of 12 super-6 ncounterers were Internet users, and only one among them stated that the Internet is her preferred environment for information encountering. The remaining seven respondents stated that the Internet is not very conducive to information encountering and expressed a preference for more traditional environments such as print and people. The super-encounterers helped identify the following barriers to information encountering on the Internet: *(1) technical barriers (e.g.,* limited access, slow response time, cumbersome navigation); *(2) information barriers (i.e.,* environment that is "too loaded" with pre-structured information); and, *(3) psychological barriers (e.g.,* a fear of becoming to obsessed with information encountering).

Another effort to address information encountering in an electronic environment was provided by Toms (1997) who conducted an experimental study that specifically focused on information encountering in digital newspapers. She

studied 47 users of daily digital newspapers by experimentally manipulating the purpose for which users approach the information system and the tools used to facilitate that access. She reported that "chance encounters revealed the *most interesting* information in a session", and that participants "recognized the need for and value" of these experiences. Toms also confirmed statements from earlier discussed studies that information encountering is enriching and rewarding to the individual.

The research on information encountering profiled in this section was conducted in the early to mid-90s. These research efforts were not specifically focused on the Web, which has emerged as a major information use environment. While these studies provide informative insight into characteristics of information encountering in general, as well as in electronic environments, we suspected that the rapid evolvement of the Web provided new and unique opportunities for information encountering, and attempted to explore this topic in a research program reported in this paper.

## CURRENT RESEARCH OF INFORMATION ENCOUNTERING ON THE WEB

In 1998 we started a research project to explore possible uses for information encountering in the context of online shopping, especially as related to customer referrals of encountered information. The conceptual framework for the analysis of information encountering (i.e. user, environment, information, and needs) that emerged from our earlier work (Erdelez, 1995) was used as a starting point. This framework: (1) informed us about differences in users' predispositions for information encountering (user dimension); (2) brought to our attention differences in information use environments in providing opportunities for information encountering (environment dimension); and (3) guided our understanding of specific types of information use, such as sharing of information which is encountered for others (information and need dimensions).

Our research started with two exploratory studies designed to eventually become parts of a larger mosaic addressing the major objective of the project. These two studies, which are reported in this paper, aimed to provide an initial answer about the characteristics of users' information encountering on the Web, in particular:

1. What are the users perceptions about the Web as an information encountering environment?
2. What do information users encounter on the Web for others and how do they share that information?

The studies relied on an exploratory research design including group interviews, individual interviews and a survey as data collection techniques. In the next section of this paper, we provide an overview of research procedures and the selection of key findings for each study. A more detailed discussion of individual studies will follow in separate papers by this author and her collaborators.

### Group interviews with super-encounterers

The research team decided to start addressing the two research questions by conducting group interviews with super-encounterers. We had hoped that people who very often experience information encountering would provide the richest source of data on information encountering on the Web. The group interviews were selected as a data collection method over individual interviews to benefit from the interactions among respondents. The respondents were recruited from the pool of super-encounterers interviewed in 1995 study of information encountering by Erdelez. Besides having established good rapport with these respondents the research team hoped to find out if there had been any changes in the respondents' attitude towards the Internet and the Web as information-encountering environments.- As reported earlier in this paper, the super-encounterers in the 1995 study did not find the Internet conducive to information encountering due to a number of technical, informational, and psychological barriers (Erdelez, 1996). Since 1995, however, we witnessed many technological advancements in accessing and processing information on the Internet, and the proliferation of the Web as an information rich environment. We suspected that these changes might have also influenced a change in the super-encounterers attitude towards information encountering on the Web.

We successfully found eight respondents from the 12 interviewed in 1995. While all eight of these respondents were enthusiastic about participating in the follow-up study only five of them were actually able to attend the group interviews due to scheduling conflicts. There were two group interviews; each lasted for about 60 minutes. To maintain a uniform structure the research team relied on a broad interview guide that prompted respondents to talk about the environments where they regularly encounter information, whether they encounter information for others,

and how they share such information. The respondents were also encouraged to talk about activities they regularly do that may facilitate information encountering.

The analysis of interview data revealed that our super-encounterers continue to dislike using the Internet and the Web for information encountering. Their time spent on the Web is rather limited (about one hour per week at most) and is used primarily for emailing and occasionally for participating in electronic discussion groups. They find the traditional environments, such as paper and personal contacts with other people, the most rewarding for encountering. Comments about the Web by several respondents in particular were very similar to the statements they provided in the 1995 study, such as: "there is too much stuff out here", "I get lost", "it will eat my time if I let it go."[11]

The analysis of the behavioral characteristics of the focus group participants revealed that the vast majority of these characteristics were also centered in a non-electronic environment. For example, respondents stated they: are avid book readers, regularly subscribe to and/or read several journal type magazines, and like to clip and save interesting information, and enjoy the act of finding and sharing information they've found with a group of their social and professional contacts. Interestingly, none of these activities occurred on the Web.

The group interviews provided us with interesting information about the lack of super-encounterers' use of the Web for information encountering, but did not provide sufficient insight into how people use the Web for encountering and sharing information relevant to others, the research question #2 for our study. Therefore, in our next research step we decided to shift our focus from super-encounterers to respondents who have unspecified information encountering habits across information environments, and who often share information encountered on the Web.

**Survey and interviews about sharing information encountered for other people on the Web**

In our initial study of information encountering (Erdelez, 1995) we relied on a combination of surveys and in-depth interviews to collect detailed descriptions of users' information encountering experiences in general. In the current study we returned to the same approach in order to, (1) identify and select the respondents who meet our study criteria in the use of Web for encountering and sharing information with others, and (2) gain in-depth understanding of processes and rationale involved in sharing information encountered for others on the Web.

We surveyed 61 students (35 undergraduate and 26 graduate) at the University of Texas at Austin about:

- the frequency of their Web and email use,
- the frequency of their encounters of information relevant to others in traditional and electronic formats,
- the frequency of their use of specific methods for sharing the information encountered for others, and
- the frequency with which information with specific content is encountered and shared with others.

After a preliminary analysis of survey data, 5 respondents with the combined highest level of Web use and the highest level of sharing encountered information were invited for follow-up interviews. The interviews, each approximately 30-40 minutes long, followed a loosely defined interview guide. The respondents were first questioned about how they encounter information for others in general and about the reasons this information is shared. Then, the respondents were asked to recall their most recent experience of encountering/sharing information on the Web. The final set of questions focused on the type of content they are likely to share on the Web, and their perceptions of electronic sharing tools.

Due to the nature of the academic population from which the study sample was drawn all the respondents were intensive Web users. 64% (39) of the respondents use the Web every day. More than 50% (32) use the Web from 2-10 hours per week. Not surprisingly, 88% of the respondents most frequently use the Web for school or work. 45% (28) of the respondents said their least frequent activity for which the Web is used is entertainment, and another 41% (25) of the respondents reported random surfing as their least frequent activity. The respondents' survey also showed a wide use of email with 81% (50) checking email at least once every day.

In response to the question about encountering information for others, 45% (28) respondents reported that they occasionally come across information that may be of use to someone else. The number of respondents who frequently have these experiences (30%, 18) was higher than the number of those who have them rarely or never (16%, 10). There was an interesting difference among the graduate and undergraduate segments of the sample

regarding the sources where information is most frequently encountered for others. The graduate students most frequently had these experiences in print resources, while undergraduates preferred personal contacts and the Web. Both groups of students, however, were in agreement regarding the most and least frequently used methods for sharing information encountered for others. The most frequently used methods of sharing were "Writing an Email" (59%,36) and "Email forwarding" (49%,30). The least frequently used methods of sharing were faxing (88%,54) and using the postal service (80%, 49). Respondents also reported that they very rarely use sharing tools provided on the Web browser (e.g., Internet Explorer's "send link by email" feature) or on the Web site itself (e.g., "email this to a friend" feature). Finally, regarding the content of information that is most frequently encountered and shared with others, graduate students shared most often entertainment and personal type information, while undergraduate students shared work/study related information. Interestingly, retail information was the second preferred choice for undergraduates and the second least preferred choice for graduate students.

The survey was followed up with face-to-face interviews with five respondents who, according to our survey data, were intensive users of the Web (with 15 or more hours of Web use per week) and who frequently encounter information for others on the Web. The interview data confirmed that respondents encounter and share information encountered for others across various information environments. These experiences on the Web are very regular; the respondents had them from "a couple a times a week" to several times a day. Respondents, however, had difficulty in recalling a particular episode of encountering information for others and the specific steps of the process involved in this activity. The prevalent method of sharing encountered information among the respondents is copying and pasting a URL link to useful information into an email message. The respondents were either not familiar with Web-based sharing tools (a.k.a. recommender forms) or had concerns about how they work. Some of these concerns involved issues such as privacy protection, security, and lack of customization to respondents needs and use patterns.

From the two studies described above we were able to gain initial insight into selected characteristics of information encountering on the Web. The following discussion will attempt to relate these findings and develop two propositions that may help broaden our knowledge of not only information encountering on the Web but also the concept of information encountering in general.

## DISCUSSION

### The framework for discussion

Our understanding of information encountering on the Web is facilitated with our parallel efforts to (1) identify the factors that influence information encountering in general and (2) study the functional components of an information encountering event. These two broad research objectives evolved from our continuing research on information encountering. In this section we first briefly describe the emerging framework for understanding information encountering in general, and then use this framework to structure our discussion of the preliminary findings on characteristics of information encountering on the Web obtained from our current research.

First, we propose that in a very broad sense the emergence of an information encountering event may be influenced by (1) the characteristics of the information users and (2) the characteristics of the information environment that provides the context for information encountering. On the users side, especially relevant to the emergence of information encountering is a user's capability to notice information that appears interesting or useful, and a user's willingness to step away from the primary task at hand, if such is present. Regarding the environment, the important features that may influence information encountering deal with the convenience that the environment provides for identifying, examining, storing, and sharing encountered information.

Furthermore, we believe that an information-encountering event can be perceived as a "divergence" from some other information-related activity or any other casual activity the user may be pursuing. We propose that one can identify the following functional components of an information-encountering episode:

- NOTICING - unexpected information catches user's attention
- STOPPING - user is temporarily diverged from some other activity that was under way and devotes attention to the information encountered
- EXAMINATION - user becomes involved in the information encountered, by reading it or exploring it in some other way
- STORING - user mentally or physically saves the information for some future use
  USE - encountered information is used for personal needs and/or shared with others
- RETURNING - user returns to the activity that was interrupted with information encountering.

For the purposes of simplification the proposed functional components focus on the behavioral aspects of an information encounter. However, a more comprehensive model of the information encountering process should also account for both the cognitive and affective aspects of a user's information behavior. This is reported in our early research (Erdelez, 1995) and is also in agreement with the work of other information-user scholars (e.g., Dervin and Nilan,1986; Kuhlthau, 1993).

**Preliminary Interpretation of Study Findings**

We started our research with an assumption that people who generally very often encounter information (super-encounterers) would be good resources for learning about information encountering on the Web. We also speculated that our early findings about the lack of interest in information encountering on the Internet by super-encounterers may be explained by the nascent nature of the Internet and the Web during our initial study conducted in 1995. However, the salient theme from the super-encounterers who participated in group interviews in our current study is that they continue to prefer traditional environments to the Web for information encountering. Their attitude towards the Web as an encountering environment did not change over the period of five years that passed since our initial study. The growth of the Web as an information repository, technological changes that improved Web accessibility, and advancements in retrieval and evaluation of information on the Web did not seem to induce super-encounterers to improve their low rating of the Web as a source for information encountering.

Why are super-encounterers in other information-use environments reluctant Web-encounterers? They are capable and willing to encounter information, and the Web offers many features conducive to encountering: hyperlinks, an abundance of information, and support for information sharing, to name just a few. To address this question, we will rely on the above-proposed framework with the functional components of an information-encountering event (noticing, stopping, examination, storing, use, and returning), and will discuss how these components pertain to the Web. Because our research is still in progress, in this paper we will only discuss selected components, i.e., "noticing", "stopping", and the sharing dimension of "use".

During the "noticing" component unexpected information gains the user's attention. By the sheer volume of information contained, the Web intuitively appears to be the ultimate place for encountering information. However, in addition to the amount of information available, presentation and organization of information may also play an important role in noticing information for encountering. While there is plenty of research-based guidance for user interface design in the context of human-computer interaction in general (see Thimbleby, 1990, for a good overview) this is an area where the Web still lacks authoritative direction. Many websites are not evaluated for usability and usefulness to actual users and are instead very often the result of a Web-designers artistic preference or a Web-developers desire to demonstrate the power of the latest Web development tools. Hopefully, the results coming from the field of hypertext research (that is being integrated into the context of the Web environment) and the efforts of popular commercially oriented web-usability gurus such as Nielsen (1999) and Spool et al., (1999) will enhance the improvement and standardization of Web interfaces.

In order for information encountering to happen, a user has to become diverted from some other activity. At the "stopping" component the user switches attention from one focus (a problem or general interest area at hand) to the information that is encountered. Hyperlinks conveniently connect users to information content that in other traditional information environments would not be easily accessible. For example, by clicking on a hyperlink, it is easy to diverge and find out more about the educational institution listed in a biographic profile of book author available on the Web. If one were reading a book, the same task would require yet another focused information search effort. Hyperlinks are a very simple but also a very powerful feature of the Web and their presence may

actually entice users to stop and explore encountered information after it has been noticed. To prevent the information encountering from occurring, a user must resist the temptation and stays focused on the primary activity underway.

The next component we will address here is the "use" component. It conceptually includes use of encountered information for personal information needs and for sharing the information encountered for other information users. Our research focused only on the sharing of information encountered with other users. As presented earlier, the Web is gaining ground as an environment where our study respondents most often encounter information for others. The most frequently used environments are still the traditional resources such as print resources and personal contacts. The traditional means of sharing information encountered for others, such as personal contact and various methods of sharing print information, are being replaced by email. It is surprising, however, that email features for sharing encountered information are only the very basic ones, e.g., copy and paste. Even the respondents who often encounter information for others on the Web do not rely on Web-based sharing tools (e.g., "send this document to a friend") even though these tools were specially designed to facilitate electronic sharing. It appears that functionality and ease of use of Web-based tools, such as sharing tools, does not by itself guarantee that users will adopt them in their regular use patterns.[2]

## Conclusion

One of the early commentators of the Web environment search tools have stated that "...serendipity is a way of finding information on the Web" (Scisco, 1995). The information encountering research suggests, however, that the serendipitous or accidental discovery of information is not some second-rate, haphazard behavior but a form of information acquisition complementary to other active and problem-specific types of information behavior. In our current research program, by building upon our earlier exploratory and descriptive research of information encountering, we hope to provide initial understanding and explanation of encountering on the Web. Based on the research completed up to this point we developed two preliminary propositions about information encountering in general that may also help illuminate the users' information encountering experiences on the Web.

*1. Information users' capability to encounter information relates to the level of their sensitivity to the information environment.*

For example, super-encounterers may be more than usually sensitive to information stimuli and thus notice more information outside the realm of their active focus then users who are generally less inclined to encounter information. Their very sensitive system of perception brings an abundance of stimuli from the environment to the super-encounterers' attention. An overly saturated information environment, such as the Web, may present "an attack" to the super-encounterers' senses, therefore causing discomfort and dissatisfaction with the environment.

*2. Information encountering is a habitual activity adjusted to the unique characteristics of each information environment.*

Super-encounterers have become accustomed to the strong and weak features of traditional information environments and have developed personal strategies for information encountering in these environments. A new information environment, such as the Web, requires users to develop new information encountering habits; different from those to which they are accustomed.

These propositions currently serve as guidelines for our future research into information encountering on the Web. These research efforts complement research by others that focus on purposeful information seeking on the Web. The importance of the Web-based information encountering research is emphasized by Toms (1998) who suggests that in our rush to design a new digital world of information centered on information filtering, intelligent agents, and topical specialization we may inversely "seriously handicap the ability of a user to experience chance encounter with digital information".

The Web provides new challenges and opportunities for information encountering. In order to "survive" on the Web super-encounterers from traditional environments may need to adjust their habitual information encountering methods to a new environment. For the majority of other users, the information density of the Web provides an opportunity to have more information encountering experiences. We hope our continuing research in this area will

uncover knowledge that will encourage super-encounterers to embrace the Web, and will help other users discover a more satisfying information encountering experience.

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## NOTES

1 The total number (N) of information encountering events analyzed in 1995 study was 132, howewer, 11 of these events did not include sufficiently specific indication of the environment where information encountering has occurred.

2 More about the use of and characteristics of Web-based sharing tools can be found in the most recent work by Erdelez & Rioux (2000).